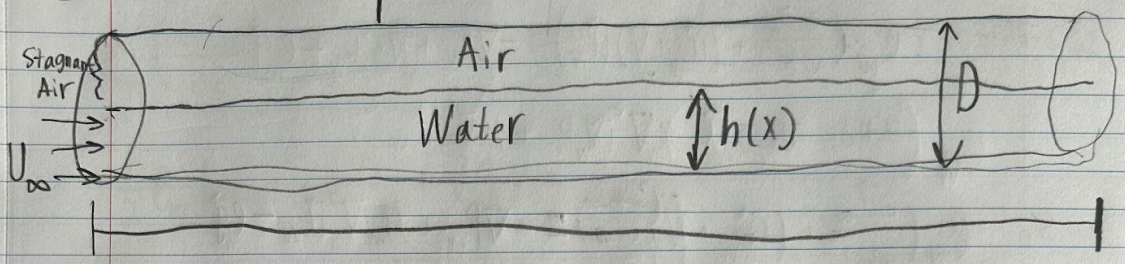
Incompressible Water Pipe Flow



$h(x) < D$ and $\delta_w(x) < h(x)$

$D \ll L$

$\therefore h(x) \ll L \quad \therefore \delta_w(x) \ll L$

Incompressible Continuity (Water)

$$\nabla \cdot \vec{u}_w = 0$$

$$\frac{\partial u_w}{\partial x} + \frac{\partial v_w}{\partial y} = 0$$

Navier-Stokes Momentum (water)

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + (\vec{u}_w \cdot \nabla)\vec{u}_w\right) = -\nabla P_w + \nabla \cdot \tau_w + \rho \vec{g}$$

$$\tau_w = \mu_w(\nabla \vec{u}_w + (\nabla \vec{u}_w)^T) + \lambda_w \cancel{(\nabla \cdot \vec{u}_w)}_0$$

$$\nabla \cdot \tau_w = \mu_w \nabla^2 \vec{u}_w + \mu \nabla \cancel{(\nabla \cdot \vec{u}_w)}_0$$

$$\nabla \cdot \tau_w = \mu_w \nabla^2 \vec{u}_w$$

Incompressible Pipe Flow

Since $\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + (\vec{u}_w \cdot \nabla)\vec{u}_w\right) = -\nabla P_w + \nabla \cdot \tau_w - \rho_w \vec{g}$

and $\nabla \cdot \tau_w = \mu_w \nabla^2 \vec{u}_w$

So $\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + (\vec{u}_w \cdot \nabla)\vec{u}_w\right) = -\nabla P_w + \mu_w \nabla^2 \vec{u}_w - \rho_w \vec{g}$

$(\vec{u}_w \cdot \nabla)\vec{u}_w = \nabla\left(\frac{1}{2}|\vec{u}_w|^2\right) + \vec{\omega}_w \times \vec{u}_w$

$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \nabla\left(\frac{1}{2}|\vec{u}_w|^2\right) + \vec{\omega}_w \times \vec{u}_w\right) = -\nabla P_w + \mu_w \nabla^2 \vec{u}_w - \rho_w \vec{g}$

$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) + \rho_w \nabla\left(\frac{1}{2}|\vec{u}_w|^2\right) = -\nabla P_w - \rho_w \vec{g} + \mu_w \nabla^2 \vec{u}_w$

$\rho_w \vec{g} = \begin{pmatrix} 0 \\ \rho_w g \end{pmatrix} \qquad \phi_g = \rho_w g y$

$\nabla \phi_g = \begin{pmatrix} 0 \\ \rho_w g \end{pmatrix} = \rho_w \vec{g}$

$\nabla(\rho_w g y) = \rho_w \vec{g}$

$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) + \nabla\left(\frac{1}{2}\rho_w |\vec{u}_w|^2\right) = -\nabla P_w - \nabla(\rho_w g y) + \mu_w \nabla^2 \vec{u}_w$

$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) = -\nabla\left(\frac{1}{2}\rho_w |\vec{u}_w|^2\right) - \nabla P_w - \nabla(\rho_w g y) + \mu_w \nabla^2 \vec{u}_w$

Incompressible Pipe Flow

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) = -\nabla\left(\frac{1}{2}\rho_w |\vec{u}_w|^2 + P_w + \rho_w g y\right) + \mu_w \nabla^2 \vec{u}_w$$

$$\frac{1}{2}\rho_w |\vec{u}_w|^2 + P_w + \rho_w g y = P_{0,w}$$

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) = -\nabla P_{0,w} + \mu_w \nabla^2 \vec{u}_w$$

$$\nabla \times \vec{u}_w = \vec{\omega}_w$$

$$\vec{\omega}_w = \nabla \times \vec{u}_w = \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ 0 \end{pmatrix} \times \begin{pmatrix} u_w \\ v_w \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y} \end{pmatrix} \text{ 2D}$$

$$\vec{\omega}_w \times \vec{u}_w = \begin{pmatrix} 0 \\ 0 \\ \frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y} \end{pmatrix} \times \begin{pmatrix} u_w \\ v_w \\ 0 \end{pmatrix} = \det \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y} \\ u_w & v_w & 0 \end{vmatrix}$$

$$\vec{\omega}_w \times \vec{u}_w = \left(\frac{\partial u_w}{\partial y} - \frac{\partial v_w}{\partial x}\right) v_w \hat{i} + \left(\frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y}\right) u_w \hat{j}$$

$$\vec{\omega}_w \times \vec{u}_w = \begin{pmatrix} \left(\frac{\partial u_w}{\partial y} - \frac{\partial v_w}{\partial x}\right) v_w \\ \left(\frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y}\right) u_w \end{pmatrix} \text{ 2D}$$

Incompressible Pipe Flow

Since

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) = -\nabla P_{o,w} + \mu_w \nabla^2 \vec{u}_w ; \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

And $\vec{\omega}_w \times \vec{u}_w = \begin{pmatrix} \left(\frac{\partial u_w}{\partial y} - \frac{\partial v_w}{\partial x}\right) v_w \\ \left(\frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y}\right) u_w \end{pmatrix}$

X-Momentum (Water)

$$\rho_w\left(\frac{\partial u_w}{\partial t} + \left(\frac{\partial u_w}{\partial y} - \frac{\partial v_w}{\partial x}\right) v_w\right) = -\frac{\partial P_{o,w}}{\partial x} + \mu_w \frac{\partial^2 u_w}{\partial x^2} + \mu_w \frac{\partial^2 u_w}{\partial y^2}$$

Nondimensionalize

$$\frac{\partial u_w}{\partial x} \sim \frac{U_\infty}{L}, \quad \frac{\partial u_w}{\partial y} \sim \frac{U_\infty}{h}, \quad \frac{\partial v_w}{\partial x} = \frac{\bar{v}_w}{L}, \text{ and } \frac{\partial v_w}{\partial y} = \frac{\bar{v}_w}{h}$$

where $\bar{v}_w$ is the spatial average y-direction velocity of water.

Therefore, the nondimensional variables: $u_w' = \frac{u_w}{U_\infty}$, $v_w' = \frac{v_w}{\bar{v}_w}$, $x' = \frac{x}{L}$, and $y' = \frac{y}{h}$

$$\rightarrow \rho_w\left(\frac{\partial u_w}{\partial t} + \left(\frac{U_\infty}{h}\frac{\partial u_w'}{\partial y'} - \frac{\bar{v}_w}{L}\frac{\partial v_w'}{\partial x'}\right) v_w\right) = -\frac{\partial P_{o,w}}{\partial x} + \mu_w \frac{U_\infty}{L^2}\frac{\partial^2 u_w'}{\partial x'^2} + \mu_w \frac{U_\infty}{h^2}\frac{\partial^2 u_w'}{\partial y'^2}$$

$$\therefore \rho_w\left(\frac{\partial u_w}{\partial t} + \left(\frac{U_\infty}{h}\frac{\partial u_w'}{\partial y'} - \frac{\bar{v}_w}{L}\frac{\partial v_w'}{\partial x'}\right) v_w\right) = -\frac{\partial P_{o,w}}{\partial x} + \mu_w \frac{U_\infty}{h^2}\left(\frac{h^2}{L^2}\frac{\partial^2 u_w'}{\partial x'^2} + \frac{\partial^2 u_w'}{\partial y'^2}\right)$$

$0,\ h \ll L$

Incompressible Pipe Flow

So,

$$\rho_w\left(\frac{\partial u_w}{\partial t}+\left(\frac{U_\infty}{h}\frac{\partial u'}{\partial y'}-\frac{\bar{v}_w}{L}\frac{\partial v_w'}{\partial x'}\right)v_w\right)=-\frac{\partial P_{o,w}}{\partial x}+\mu_w\frac{U_\infty}{h^2}\frac{\partial^2 u_w'}{\partial y'^2}$$

Redimensionalize right-side of equation.

$$\therefore \rho_w\left(\frac{\partial u_w}{\partial t}+\left(\frac{U_\infty}{h}\frac{\partial u'}{\partial y'}-\frac{\bar{v}_w}{L}\frac{\partial v_w'}{\partial x'}\right)v_w\right)=-\frac{\partial P_{o,w}}{\partial x}+\mu_w\frac{\partial^2 u_w}{\partial y^2}$$

Incompressible Continuity

$$\frac{\partial u_w}{\partial x}+\frac{\partial v_w}{\partial y}=0$$

Nondimensionalize

$$\frac{U_\infty}{L}\frac{\partial u_w'}{\partial x'}+\frac{\bar{v}_w}{h}\frac{\partial v_w'}{\partial y'}=0$$

$$\frac{h}{U_\infty}\left(\frac{U_\infty}{L}\frac{\partial u_w'}{\partial x'}+\frac{\bar{v}_w}{h}\frac{\partial v_w'}{\partial y'}\right)=\left(\frac{h}{U_\infty}\right)(0)$$

$$\frac{h}{L}\frac{\partial u_w'}{\partial x'}+\frac{\bar{v}_w}{U_\infty}\frac{\partial v_w'}{\partial y'}=0$$

$h \ll L$

$$\frac{\bar{v}_w}{U_\infty}\frac{\partial v_w'}{\partial y'}=0 \text{ and } \frac{\partial v_w'}{\partial y'}\sim O(1) \therefore \frac{\bar{v}_w}{U_\infty}O(1)\cong 0$$

Incompressible Pipe Flow

Since $\frac{\bar{V}_w}{U_\infty} O(1) \cong 0$, $\frac{\bar{V}_w}{U_\infty} \cong 0$, $\therefore \bar{V}_w \ll U_\infty$

$$\bar{V}_w \cong 0$$

And $$\rho_w \left(\frac{\partial u_w}{\partial t} + \left(\frac{U_\infty}{h} \frac{\partial u'}{\partial y'} - \frac{\bar{V}_w}{L} \frac{\partial v'_w}{\partial x'} \right) V_w \right) = - \frac{\partial P_{o,w}}{\partial x} + \mu_w \frac{\partial^2 u}{\partial y^2}$$

$$\therefore \rho_w \left(\frac{\partial u_w}{\partial t} + \left(\frac{U_\infty}{h} \right) \left(\frac{\partial u'}{\partial y'} - \frac{h \bar{V}_w}{L U_\infty} \frac{\partial v'_w}{\partial x'} \right) V_w \right) = - \frac{\partial P_{o,w}}{\partial x} + \mu_w \frac{\partial^2 u}{\partial y^2}$$

0, $h \ll L$ and $\bar{V}_w \ll U_\infty$

$$\rho_w \left(\frac{\partial u_w}{\partial t} + \frac{U_\infty}{h} \frac{\partial u'}{\partial y'} V_w \right) = - \frac{\partial P_{o,w}}{\partial x} + \mu_w \frac{\partial^2 u}{\partial y^2}$$

Redimensionalize left side of the equation

$$\rho_w \left(\frac{\partial u_w}{\partial t} + V_w \frac{\partial u}{\partial y} \right) = - \frac{\partial P_{o,w}}{\partial x} + \mu_w \frac{\partial^2 u}{\partial y^2} \quad (1)$$

Incompressible Pipe Flow

Since $\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) = -\nabla P_{0,w} + \mu_w \nabla^2 \vec{u}_w$

And $\vec{\omega}_w \times \vec{u}_w = \begin{pmatrix} \left(\frac{\partial u_w}{\partial y} - \frac{\partial v_w}{\partial x}\right) v_w \\ \left(\frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y}\right) u_w \end{pmatrix}$

Y-momentum (Water)

$$\rho_w\left(\frac{\partial v_w}{\partial t} + \left(\frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y}\right) u_w\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w \frac{\partial^2 v_w}{\partial x^2} + \mu_w \frac{\partial^2 v_w}{\partial y^2}$$

Nondimensionalize

$$\rho_w\left(\frac{\partial v_w}{\partial t} + \left(\frac{\bar{v}_w}{L}\frac{\partial v_w'}{\partial x'} - \frac{U_\infty}{h}\frac{\partial u_w'}{\partial y'}\right) u_w\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w \frac{\bar{v}_w}{L^2}\frac{\partial^2 v_w'}{\partial x'^2} + \mu_w \frac{\bar{v}_w}{h^2}\frac{\partial^2 v_w'}{\partial y'^2}$$

$$\rho_w\left(\frac{\partial v_w}{\partial t} + \left(\frac{U_\infty}{h}\right)\left(\frac{h}{L}\frac{\bar{v}_w}{U_\infty}\frac{\partial v_w'}{\partial x'} - \frac{\partial u_w'}{\partial y'}\right) u_w\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w \frac{\bar{v}_w}{h^2}\left(\frac{h^2}{L^2}\frac{\partial^2 v_w'}{\partial x'^2} + \frac{\partial^2 v_w'}{\partial y'^2}\right)$$

$h \ll L$ and $\bar{v}_w \ll U_\infty$ (the $\frac{h}{L}\frac{\bar{v}_w}{U_\infty}$ term $\to 0$; $\frac{h^2}{L^2} \to 0$ since $h \ll L$)

$$\rho_w\left(\frac{\partial v_w}{\partial t} - \frac{U_\infty}{h}\frac{\partial u_w'}{\partial y'} u_w\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w \frac{\bar{v}_w}{h^2}\frac{\partial^2 v_w'}{\partial y'^2}$$

Redimensionalize

$$\rho_w\left(\frac{\partial v_w}{\partial t} - u_w \frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w \frac{\partial^2 v_w}{\partial y^2}$$

Incompressible Pipe Flow

$$\rho_w\left(\frac{\partial v_w}{\partial t} - u_w\frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_{o,w}}{\partial y} + \mu_w\frac{\partial^2 v_w}{\partial y^2}$$

$$\rho_w\left(\frac{\partial v_w}{\partial t} - \frac{\partial\left(\frac{1}{2}u_w^2\right)}{\partial y}\right) = -\frac{\partial P_{o,w}}{\partial y} + \mu_w\frac{\partial^2 v_w}{\partial y^2}$$

$$P_{o,w} = P_w + \frac{1}{2}\rho_w|\vec{v}_w|^2 + \rho_w g y$$

$$\therefore P_{o,w} = P_w + \frac{1}{2}\rho_w u_w^2 + \frac{1}{2}\rho_w v_w^2 + \rho_w g y$$

$$\rho_w\frac{\partial v_w}{\partial t} - \rho_w\frac{\partial\left(\frac{1}{2}u_w^2\right)}{\partial y} = -\frac{\partial\left(P_w + \frac{1}{2}\rho_w u_w^2 + \frac{1}{2}\rho_w v_w^2 + \rho_w g y\right)}{\partial y} + \mu\frac{\partial^2 v_w}{\partial y^2}$$

$$\rho_w\frac{\partial v_w}{\partial t} - \cancel{\frac{\partial\left(\frac{1}{2}\rho_w u_w^2\right)}{\partial y}} = -\frac{\partial\left(P_w + \frac{1}{2}\rho_w v_w^2 + \rho_w g y\right)}{\partial y} - \cancel{\frac{\partial\left(\frac{1}{2}\rho_w u_w^2\right)}{\partial y}} + \mu_w\frac{\partial^2 v}{\partial y^2}$$

$$\rho_w\frac{\partial v_w}{\partial t} = -\frac{\partial\left(P_w + \frac{1}{2}\rho_w v_w^2 + \rho_w g y\right)}{\partial y} + \mu_w\frac{\partial^2 v}{\partial y^2}$$

$P_w + \frac{1}{2}\rho_w v_w^2 + \rho_w g y$, nondimensionalize the expression

$$P_w' = \frac{P_w - P_\infty}{\frac{1}{2}\rho_w U_\infty^2}, \quad v_w' = \frac{v_w}{\bar{v}_w} \quad \therefore \frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty + \frac{1}{2}\rho_w \bar{v}_w^2 v_w'^2 + \rho_w g y$$

$$\Rightarrow P_w = \frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty \text{ and } v_w = \bar{v}_w v_w'$$

Incompressible Pipe Flow

Approximate the expression

$$\frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty + \frac{1}{2}\rho_w \bar{V}_w^2 V_w'^2 + \rho_w g y$$

$$\frac{1}{2}\rho_w U_\infty^2 \left(P_w' + \frac{\frac{1}{2}\rho_w \bar{V}_w^2}{\frac{1}{2}\rho_w U_\infty^2}(V_w')^2\right) + P_\infty + \rho_w g y$$

$$\frac{1}{2}\rho_w U_\infty^2 \left(P_w' + \frac{\bar{V}_w^2}{U_\infty^2}(V_w')^2\right) + P_\infty + \rho_w g y$$

$0, \bar{V}_w \ll U_\infty$

$$\therefore \frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty + \rho_w g y$$

$P_w + \rho g y$ is now the expression

So $$\rho_w \frac{\partial v_w}{\partial t} = -\frac{\partial (P_w + \frac{1}{2}\rho_w V_w^2 + \rho_w g y)}{\partial y} + \mu_w \frac{\partial^2 v_w}{\partial y^2}$$

$$\cong \rho_w \frac{\partial v_w}{\partial t} = -\frac{\partial (P_w + \rho_w g y)}{\partial y} + \mu \frac{\partial^2 v_w}{\partial y^2}$$

$$\rho_w \frac{\partial v_w}{\partial t} = -\frac{\partial P_w}{\partial y} - \rho g_w + \mu \frac{\partial^2 v_w}{\partial y^2}$$

$$\rho_w \frac{\partial v_w}{\partial t} + \frac{\partial P_w}{\partial y} + \rho_w g = \mu \frac{\partial^2 v_w}{\partial y^2} \quad (2)$$

Incompressible Pipe Flow

(1) $$\rho_w\left(\frac{\partial u_w}{\partial t} + v_w\frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_{0,w}}{\partial x} + \mu_w\frac{\partial^2 u_w}{\partial y^2}$$

(2) $$\rho_w\frac{\partial v_w}{\partial t} + \frac{\partial P_w}{\partial y} + \rho_w g = \mu_w\frac{\partial^2 v_w}{\partial y^2}$$

Where $$P_{0,w} = \frac{1}{2}\rho_w|\vec{u}_w|^2 + \rho_w g y + P_w$$

$$P_{0,w} = \frac{1}{2}\rho_w(u_w^2 + v_w^2) + P_w + \rho_w g y$$

$$\frac{\bar{v}_w}{U_\infty} \sim \frac{h}{L}$$

$$P_{0,w} = \frac{1}{2}\rho_w U_\infty^2\left(u_w'^2 + \left(\frac{\bar{v}_w}{U_\infty}\right)^2 v_w'^2\right) + P_w + \rho_w g y$$

$0, \bar{v}_w \ll U_\infty$

$$P_{0,w} = \frac{1}{2}\rho_w U_\infty^2 u_w'^2 + P_w + \rho_w g y$$

$$P_{0,w} = \frac{1}{2}\rho_w u_w^2 + P_w + \rho_w g y$$

$$\therefore \frac{\partial P_{0,w}}{\partial x} = \frac{\partial\left(\frac{1}{2}\rho_w u_w^2 + P_w + \rho_w g y\right)}{\partial x} = \rho_w u_w\frac{\partial u_w}{\partial x} + \frac{\partial P_w}{\partial x}$$

$$-\frac{\partial P_{0,w}}{\partial x} = -\rho_w u_w\frac{\partial u_w}{\partial x} - \frac{\partial P_w}{\partial x}$$

Incompressible Pipe Flow

Since,

$$\rho_w\left(\frac{\partial u_w}{\partial t}+V_w\frac{\partial u_w}{\partial y}\right)=-\frac{\partial P_{o,w}}{\partial x}+\mu_w\frac{\partial^2 u_w}{\partial y^2}$$

$$\rho_w\left(\frac{\partial u_w}{\partial t}+V_w\frac{\partial u_w}{\partial y}\right)=-\rho_w U_w\frac{\partial u_w}{\partial x}-\frac{\partial P_w}{\partial x}+\mu_w\frac{\partial^2 u_w}{\partial y^2}$$

$$\rho_w\left(\frac{\partial u_w}{\partial t}+u_w\frac{\partial u_w}{\partial x}+V_w\frac{\partial u_w}{\partial y}\right)=-\frac{\partial P_w}{\partial x}+\mu_w\frac{\partial^2 u_w}{\partial y^2}\quad (1)$$

$$\rho_w\frac{\partial V_w}{\partial t}+\frac{\partial P_w}{\partial y}+\rho_w g=\mu_w\frac{\partial^2 V_w}{\partial y^2}\quad (2)$$

$$\int_0^y \rho_w\frac{\partial V_w}{\partial t}\partial\bar{y}+\int_0^y\frac{\partial P_w}{\partial\bar{y}}\partial\bar{y}+\int_0^y\rho_w g\,\partial\bar{y}=\int_0^y\mu_w\frac{\partial^2 V_w}{\partial\bar{y}^2}\partial\bar{y}$$

$$\int_0^y \rho_w\frac{\partial V_w}{\partial t}V_w\,dt+P_w-P_w\Big|_{y=0}+\rho_w g y=\mu\frac{\partial V_w}{\partial y}-\mu\frac{\partial V_w}{\partial y}\Big|_{y=0}$$

Continuity (Water): $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$,

At wall, $y=0$

$\frac{\partial u}{\partial x}\Big|_{y=0}+\frac{\partial v}{\partial y}\Big|_{y=0}=0$, $u(x,y=0)=0$, no slip

$\frac{\partial}{\partial x}(u(x,y=0))=\frac{\partial}{\partial x}(0)$

$\frac{\partial u}{\partial x}\Big|_{y=0}=0$

$\frac{\partial v}{\partial y}\Big|_{y=0}=0$

Incompressible Pipe Flow

$$\int_0^y \rho_w \frac{\partial v_w}{\partial t} v_w \partial t + P_w - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v}{\partial y}$$

$$\int_0^y \rho_w v_w \frac{\partial v_w}{\partial t} \partial t + P_w - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$$\int_0^y \rho_w v_w \partial v_w + P_w - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$$\frac{1}{2}\rho_w v_w^2 - \cancel{\frac{1}{2}\rho_w v_w^2|_{y=0}} + P_w - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$v_w|_{y=0} = 0$, due to no permeability

$$\frac{1}{2}\rho_w v_w^2 + P_w - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$$P_w' = \frac{P_w - P_\infty}{\frac{1}{2}\rho_w U_\infty^2}, \quad v_w' = \frac{v_w}{\bar{v}_w}$$

$$P_w = \frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty \quad \text{and} \quad v_w = \bar{v}_w v_w'$$

$$\frac{1}{2}\rho_w \bar{v}_w^2 v_w'^2 + \frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v}{\partial y}$$

Incompressible Pipe Flow

$$\frac{1}{2}\rho_w \bar{v}_w^2 (v_w')^2 + \frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty - P_w\big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$$\frac{1}{2}\rho_w U_\infty^2 \left(\frac{\bar{v}_w^2}{U_\infty^2}(v_w')^2 + P_w' \right) + P_\infty - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$\bar{v}_w \ll U_\infty$

$$\underbrace{\frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty}_{P_w} - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$$\therefore P_w - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y} \quad (2)$$

$$\rho_w \left(\frac{\partial u_w}{\partial t} + u_w \frac{\partial u_w}{\partial x} + v_w \frac{\partial u_y}{\partial y} \right) = -\frac{\partial P_w}{\partial x} + \mu_w \frac{\partial^2 u_w}{\partial y^2} \quad (1)$$

With $P_w - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$

$$\frac{\partial}{\partial x}\left(P_w - P_w|_{y=0} + \rho_w g y \right) = \frac{\partial}{\partial x}\left(\mu_w \frac{\partial v_w}{\partial y} \right)$$

$$\therefore \frac{\partial P_w}{\partial x} - \frac{\partial P_w}{\partial x}\bigg|_{y=0} = \mu_w \frac{\partial^2 v_w}{\partial x \partial y} \; ; \; \frac{\partial u_w}{\partial x} + \frac{\partial v_w}{\partial y} = 0$$

$$\frac{\partial}{\partial x}\left(\frac{\partial u_w}{\partial x} + \frac{\partial v_w}{\partial y} \right) = \frac{\partial}{\partial x}(0)$$

$$\frac{\partial P_w}{\partial x} - \frac{\partial P_w}{\partial x}\bigg|_{y=0} = \mu_w \left(-\frac{\partial^2 u_w}{\partial x^2} \right)$$

$$\frac{\partial^2 u_w}{\partial x^2} + \frac{\partial^2 v_w}{\partial x \partial y} = 0, \; \frac{\partial^2 v_w}{\partial x \partial y} = -\frac{\partial^2 u_w}{\partial x^2}$$

$$\frac{\partial P_w}{\partial x} - \frac{\partial P_w}{\partial x}\bigg|_{y=0} = -\mu_w \frac{\partial^2 u_w}{\partial x^2} \quad (3)$$

Incompressible Pipe Flow

So $$\frac{\partial P_w}{\partial x} - \frac{\partial P_w}{\partial x}\Big|_{y=0} = -\mu_w \frac{\partial^2 u_w}{\partial x^2} \quad (3)$$

$$-\rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + V_w\frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_w}{\partial x} + \mu_w\frac{\partial^2 u_w}{\partial y^2} \quad (1)$$

At wall, $y=0$,

$$\rho_w\left(\cancel{\frac{\partial u_w}{\partial t}}^{0}\Big|_{y=0} + \cancel{u_w\frac{\partial u_w}{\partial x}}^{0} + \cancel{V_w\frac{\partial u_w}{\partial y}}^{0}\right) = -\frac{\partial P_w}{\partial x}\Big|_{y=0} + \mu_w\frac{\partial^2 u_w}{\partial y^2}\Big|_{y=0}$$

$u_w = 0$, no slip

$u_w = 0$, no slip

$V_w = 0$, no permeability

$$0 = -\frac{\partial P_w}{\partial x}\Big|_{y=0} + \mu_w\frac{\partial^2 u_w}{\partial y^2}\Big|_{y=0}$$

$$\frac{\partial P_w}{\partial x}\Big|_{y=0} = \mu_w\frac{\partial^2 u_w}{\partial y^2}\Big|_{y=0}$$

$$\rightarrow \frac{\partial P_w}{\partial x} - \mu_w\frac{\partial^2 u_w}{\partial y^2}\Big|_{y=0} = -\mu_w\frac{\partial^2 u_w}{\partial x^2}$$

$$\rightarrow \frac{\partial P_w}{\partial x} = \mu_w\frac{\partial^2 u}{\partial y^2} - \rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + V_w\frac{\partial u_w}{\partial y}\right)$$

$\therefore$ Substitute,

$$\therefore \mu_w\frac{\partial^2 u}{\partial y^2} - \rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + V_w\frac{\partial u_w}{\partial y}\right) - \mu_w\frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = -\mu_w\frac{\partial^2 u_w}{\partial x^2}$$

Incompressible Pipe Flow

So, $\mu_w \frac{\partial^2 u_w}{\partial y^2} - \rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + v_w\frac{\partial u_w}{\partial y}\right) - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = -\mu_w \frac{\partial^2 u_w}{\partial x^2}$

$$\mu_w \frac{\partial^2 u_w}{\partial y^2} + \mu_w \frac{\partial^2 u_w}{\partial x^2} - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = \rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + v_w\frac{\partial u_w}{\partial y}\right)$$

Nondimensionalize left-side of equation

$$\mu_w \frac{U_\infty}{h^2}\frac{\partial^2 u'_w}{\partial y'^2} + \mu_w \frac{U_\infty}{L^2}\frac{\partial^2 u'_w}{\partial x'^2} - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = \rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + v_w\frac{\partial u_w}{\partial y}\right)$$

$$\mu_w \frac{U_\infty}{h^2}\left(\frac{\partial^2 u'_w}{\partial y'^2} + \frac{h^2}{L^2}\frac{\partial^2 u'_w}{\partial x'^2}\right) - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = \rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + v_w\frac{\partial u_w}{\partial y}\right)$$

$0, h \ll L$

$$\mu_w \frac{U_\infty}{h^2}\frac{\partial^2 u'_w}{\partial y'^2} - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = \rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + v_w\frac{\partial u_w}{\partial y}\right)$$

$$\mu_w \frac{\partial^2 u_w}{\partial y^2} - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = \rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + v_w\frac{\partial u_w}{\partial y}\right)$$

$$\rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + v_w\frac{\partial u}{\partial y}\right) = \mu_w\frac{\partial^2 u_w}{\partial y^2} - \mu_w\frac{\partial^2 u_w}{\partial y^2}\Big|_{y=0} \quad (1)$$

$$P_w - P_w|_{y=0} + \rho_w g y = \mu_w \frac{\partial v}{\partial y} \quad (2)$$

Incompressible Pipe Flow

Since,

$$\rho_w\left(\frac{\partial u_w}{\partial t}+u_w\frac{\partial u_w}{\partial x}+v_w\frac{\partial u_w}{\partial y}\right)=\mu_w\frac{\partial^2 u_w}{\partial y^2}-\mu_w\frac{\partial^2 u_w}{\partial y^2}\bigg|_{y=0}$$

$$P_w-P_w|_{y=0}+\rho_w g y=\mu_w\frac{\partial v_w}{\partial y}$$

Continuity (Incompressible)

$$\frac{\partial u_w}{\partial x}+\frac{\partial v_w}{\partial y}=0$$

$$\frac{\partial v_w}{\partial y}=-\frac{\partial u_w}{\partial x}$$

$$P_w-P_w|_{y=0}+\rho_w g y=-\mu_w\frac{\partial u_w}{\partial x}$$

$$\therefore \frac{\partial u_w}{\partial x}=\frac{1}{\mu_w}\left(P_w|_{y=0}-P_w-\rho_w g y\right)$$

$$\rho_w\left(\frac{\partial u_w}{\partial t}+\frac{u_w}{\mu_w}\left(P_w|_{y=0}-P_w-\rho_w g y\right)+v_w\frac{\partial u_w}{\partial y}\right)=\mu_w\frac{\partial^2 u_w}{\partial y^2}-\mu_w\frac{\partial^2 u_w}{\partial y^2}\bigg|_{y=0}$$

$$P_w-P_w|_{y=0}+\rho_w g y=\mu_w\frac{\partial v_w}{\partial y}\ ;\ \text{At}\ \left(x=0, y=\frac{h(0)}{2}\right), P_w=P_\infty$$